POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Data Communication

Semester: Fall

Year : 2024
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain the block diagram of a data communication system with reference to its core components. Compare digital and analog data transmission in terms of signal processing, error handling, and bandwidth. 8

   b) Define Baud rate. An analog signal has a bit rate of 800 bps and a baud rate of 1000 baud. How many data elements are carried by each signal element? How many signal elements do we need? 7

2. a) Differentiate between energy and power signal with example. Justify whether $x(t) = e^{-a|t|}$, a>0 is energy or power signal. 7

   b) Find the Fourier series coefficient for the given signals: 8

   $x(t)=4+\sin(\omega t) + 2\cos(2\omega t) + \cos(2\omega t + \frac{\pi}{4})$ and also plot magnitude and phase spectrum.

3. a) Define signum signal and unit impulse signal. Find the Fourier Transform of Signum signal and unit step signal. 8

   b) Explain the concept of a protocol stack. Compare the TCP/IP and OSI models with respect to their approach to organizing communication protocols. 7

4. a) What are guided and unguided transmission media? Discuss the advantages of fiber optic over other guided media. 7

   b) A bit stream 1101011011 is transmitted using the cyclic redundancy check (CRC) method. The generator polynomial is $x^4 + x + 1$. Determine the transmitted codeword. 8

5. a) Explain how sender-receiver synchronization issues are addressed in the sliding window protocol, focusing on Selective Repeat mechanisms. Discuss their advantages, disadvantages, and how they ensure reliable data transmission in high-speed networks. 8

   b) Explain the significance of switching? Also, Explain FDM hierarchy in telephone system. 7

6. a) Represent the given sequence of bits 10110100 using 7
   i. Unipolar Nonreturn-to-zero (NRZ)
   ii. Polar NRZ
   iii. Bipolar encoding
   iv. Manchester encoding

   b) Differentiate Frequency Modulation (FM) and Phase Modulation (PM). Why is FM superior over AM in communication? 8

7. Write short notes on: **(Any two)** 2×5
   a) Satellite Communication
   b) Analog Hierarchy
   c) QPSK Modem

POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2024
Programme: BE | | Full Marks : 100
Course: Data Communication (New) | | Pass Marks : 45
| | Time : 3 hrs.

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Attempt all the questions.

1. a) Explain the five essential components of a data communication system. How does each component contribute to successful communication? Explain with suitable diagram. 7

   b) What is serial and parallel data transmission? Differentiate between synchronous and asynchronous data transmission. 8

2. a) Check whether the given system is linear, time invariant and causal or not. 7

   i. $y(t) = \log x(t)$

   ii. $y(t) = t^2 x(t)$

   b) Differentiate between energy and power signal. Calculate whether ramp signal is power signal or energy. 8

3. a) Why Frame Relay is faster than X.25? Explain TCP/IP layers and compare it with OSI reference model. 8

   b) What is the purpose of cladding in an optical fibre? Explain the modes of propagation used in the optical fibre. 7

4. a) Define transmission impairments. And explain the causes of transmission impairments in details. 7

   b) If the 7 bit Hamming codeword received by a receiver is 1011011. Assuming the even parity, state whether the received codeword is correct or wrong. If wrong, decode the correct data word and locate the bit in error. 8

   OR

   Compress "Engineering" using Huffman codes and find its efficiency and redundancy.

5. a) What is ARQ? What are the two types of sliding window ARQ error control? Explain. 8

   b) What do you mean by multiplexing? How is TDM different from FDM? 7

6. a) Why do we need modulation need in data communication? Explain. 7

   OR

   Calculate the rate of a 500 baud 8-QAM signal with a constellation diagram.

   b) Represent the given bit stream 10011100001101 using 8

   i. Polar NRZ

   ii. Differential Manchester

   iii. HDB3

7. Write short notes on: **(Any two)** 2×5

   a) Standards organizations

   b) Noise and its types

   c) HDLC protocol vs Point-to-Point protocol

POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2025
Programme: BE | Full Marks : 100
Course: Data Communication | Pass Marks : 45
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Candidates are required to give their answers in their own words as far as practicable.
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Attempt all the questions.

1. a) How data is transferred from sender to receiver in digital communication. Explain with necessary components. 7
   b) Discuss about serial data communication. A system sends a signal that can assume 8 different voltage levels. It sends 400 of these signals per second. What are the baud and bit rates? 8
2. a) Define impulse and ramp signal with mathematical expression. Find the energy and power signal $x(t)=e^{-4t}.\ u(t)$ and justify whether it is energy signal or power signal. 8
   b) Distinguish between continuous and discrete time system. Explain Basic system properties as Linearity, Causality, Stability, Static & Dynamic. 7
3. a) Find the Fourier transform of unit step function. Why mathematical tool like Fourier series and Fourier transform are used in engineering. 7
   b) Why Frame Relay is faster than X.25? Explain TCP/IP layers and compare it with OSI reference model. 8
4. a) Why do we need to twist the wires? Discuss the role of antennas in facilitating unguided transmission within mobile networks. 7
   b) What is attenuation? If the frame is $x^7 + x^5 + 1$ by the generator polynomial $x^3 + 1$, what would be the transmitted frame using CRC? 8
5. a) When is the retransmission necessary? Describe the two types of sliding window ARQ protocols and explain how they ensure reliable data transmission in noisy communication channels. 7
   b) Compare and contrast circuit switching, packet switching, virtual circuit packet switching. 8

6. a) Encode 01001101010110 using Manchester, AMI, NRZ, NRZ-I and explain line coding techniques. 8
   b) What is multi level modulation? What is the importance of MODEM in digital communication system? 7
7. Write short notes on: **(Any two)** 2×5
   a) Optical fiber communication
   b) Digital hierarchy in telephone system
   c) Standard organizations

POKHARA UNIVERSITY

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Attempt all the questions.

1. a) What is data communication system? Explain the block diagram of data communication. 7

   b) Define system. Explain basic system properties with its application and example. 8

2. a) Explain different types of elementary signals with its waveform and mathematical representation. 8

   b) Explain the operational principles of the ALOHA protocol in data communication. Compare it with CSMA/CD in terms of collision handling. 7

3. a) Define UTP & STP. Explain the working principle of optical fiber. Discuss the advantages and disadvantages of fiber optics. 8

   b) Explain various transmission impairments of a communication system. 7

4. a) How collision can be detected and corrected in data transmission? Explain brief the flow chart about CSMA/CD. 8

   OR

   A bit stream 11010101 is transmitted using standard CRC method. The generator polynomial is x^3+1.Show the actual bit transmitted. Suppose, the third bit from the left is inverted during transmission. Show that this error is detected at the receiver's end.

   b) What is ARQ? How does stop and wait ARQ and sliding window ARQ deals with error. 7

   OR

   Describe flow control and error control. How does Go-back –N ARQ work?

5. a) What do you understand by PBX? Compare among circuit switching, message switching and packet switching. 8

   b) Explain FDM. State sampling theorem and why we need modulation in communication system? 7

6. a) Define line coding. Explain the block diagram of PCM system. 7

   b) Encode digital data 11010101011101 using 8

   i) Unipolar RZ ii) Bipolar NRZ iii) AMI iv) Manchester

7. Write short notes on: **(Any two)** 2×5

   a) Shannon Capacity Theorem

   b) Lossy compression

   c) Point-to-Point protocol

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Attempt all the questions.

1. a) What are standards? Explain its types. Also explain the standard organizations in data communication. 8
   b) Differentiate between different types of transmission modes. Briefly explain RS-232C. 7
2. a) Define Time Invariant System. Explain the following properties of the systems: 8
      i. Causality
      ii. Linearity
      iii. Stability
      iv. Memory
   b) Explain the existence of Fourier Transform. Find the Fourier Transform of Signum function. 7
3. a) OSI layer architecture is a reference model. Support this statement and explain the basic function of each layer. 7
   b) Describe X.25, Frame Relay and ATM briefly. 8
4. a) What are guided and unguided transmission media? Discuss the advantages of fiber optics over other guided media. 7
   b) A bit stream 1101011011 is transmitted using the cyclic redundancy check (CRC) method. The generator polynomial is $x^4 + x + 1$. Determine the transmitted codeword. 8
5. a) Why flow control is needed? Explain sliding window flow control algorithm in detail. 8
   b) Explain the multiplexing technique applied in digital telephony with their significance, application and multiplexing hierarchy. 7
6. a) How modulation helps to reduce the size of antenna, illustrate in brief. Explain about, amplitude. frequency and phase modulation. 7

   b) How an analog signal is converted into digital? Explain QPSK as an multilevel modulation with necessary diagram. 8
7. **Write short notes on: (Any two)** 2×5
   a) Circuit switching and packet Switching
   b) PCM
   c) Energy and power signal

POKHARA UNIVERSITY

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Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) How data is transferred from sender to receiver in digital communication. Explain with necessary components. 7

   b) Define Baud rate. If a communication channel has a bandwidth of 3 kHz and a SNR of 31 dB, what would be the channel capacity using Shannon capacity theorem? 8

OR

Explain transmission modes in data communication.

2. a) Define LTI systems. Explain with examples, how periodic and non-periodic signals are distinguished. 8

   b) OSI layer architecture is a reference model. Support this statement and explain the basic function of each layer. 7

3. a) Describe Aloha with its types. Explain in brief the flow chart of CSMA/CD. 8

   b) Briefly explain various electromagnetic spectrum for telecommunication. 7

4. a) Explain different signal impairments with their preventive measures and necessary figures. 8

   b) Discuss the concept of redundancy in error detection and correction. Explain error correction techniques with the example of Hamming code. 7

OR

Generate the Huffman code of words "GenZ Movement" and calculate average length.

5. a) What is an error control Technique. Explain sliding window error control protocol in details. 8

   b) What do you mean by Multiplexing? Explain its types with its application in Digital Multiplexing. 7

6. a) Differentiate encoding and modulation. The sequence of the bit is 010111100. What will be the output if the following scheme is used. 8
   i. AMI
   ii. NRZ-I
   iii. NRZ-L
   iv. RZ

   b) Differentiate Frequency Modulation (FM) and Phase Modulation (PM). Between AM and FM, which one has better noise immunity? Explain. 7

7. Write short notes on: **(Any two)** 2×5
   a) HDLC Protocol
   b) Shannon Capacity Theorem
   c) Cellular Telephony